

MATHEMATICS

JEE (MAIN+ADVANCED)

NURTURE COURSE

$$M_{2 \times 2} = \begin{bmatrix} a & b \\ c & d \end{bmatrix}$$
$$D_{2 \times 2} = ad - bc$$

Exercise

Determinant

(English Medium)

EXERCISE (O-1)

Straight Objective Type

1. $\begin{vmatrix} y+z & x & x \\ y & z+x & y \\ z & z & x+y \end{vmatrix}$ equals -

(A) $x^2y^2z^2$

(B) $4x^2y^2z^2$

(C) xyz

(D) $4xyz$

DT0001

2. If $\begin{vmatrix} 1 & 3 & 4 \\ 1 & x-1 & 2x+2 \\ 2 & 5 & 9 \end{vmatrix} = 0$, then x is equal to-

(A) 2

(B) 1

(C) 4

(D) 0

DT0002

3. If $px^4 + qx^3 + rx^2 + sx + t = \begin{vmatrix} x^2+3x & x-1 & x+3 \\ x+1 & 2-x & x-3 \\ x-3 & x+4 & 3x \end{vmatrix}$ then t is equal to -

(A) 33

(B) 0

(C) 21

(D) none

DT0003

4. There are two numbers x making the value of the determinant $\begin{vmatrix} 1 & -2 & 5 \\ 2 & x & -1 \\ 0 & 4 & 2x \end{vmatrix}$ equal to 86. The sum of

these two numbers, is-

(A) -4

(B) 5

(C) -3

(D) 9

DT0004

5. If $\Delta = \begin{vmatrix} a_1 & b_1 & c_1 \\ a_2 & b_2 & c_2 \\ a_3 & b_3 & c_3 \end{vmatrix}$ and A_2, B_2, C_2 are respectively cofactors of a_2, b_2, c_2 then $a_1A_2 + b_1B_2 + c_1C_2$ is

equal to-

(A) $-\Delta$

(B) 0

(C) Δ

(D) none of these

DT0005

6. If in the determinant $\Delta = \begin{vmatrix} a_1 & b_1 & c_1 \\ a_2 & b_2 & c_2 \\ a_3 & b_3 & c_3 \end{vmatrix}$, A_1, B_1, C_1 etc. be the co-factors of a_1, b_1, c_1 etc., then which

of the following relations is incorrect-

(A) $a_1A_1 + b_1B_1 + c_1C_1 = \Delta$

(B) $a_2A_2 + b_2B_2 + c_2C_2 = \Delta$

(C) $a_3A_3 + b_3B_3 + c_3C_3 = \Delta$

(D) $a_1A_2 + b_1B_2 + c_1C_2 = \Delta$

DT0006

7. If $\Delta = \begin{vmatrix} a_1 & b_1 & c_1 \\ a_2 & b_2 & c_2 \\ a_3 & b_3 & c_3 \end{vmatrix}$ and A_1, B_1, C_1 denote the co-factors of a_1, b_1, c_1 respectively, then the value of

the determinant $\begin{vmatrix} A_1 & B_1 & C_1 \\ A_2 & B_2 & C_2 \\ A_3 & B_3 & C_3 \end{vmatrix}$ is -

- (A) Δ (B) Δ^2 (C) Δ^3 (D) 0

DT0007

8. If a, b, c are in AP, then $\begin{vmatrix} x+1 & x+2 & x+a \\ x+2 & x+3 & x+b \\ x+3 & x+4 & x+c \end{vmatrix}$ equals -

- (A) $a + b + c$ (B) $x + a + b + c$ (C) 0 (D) none of these

DT0008

9. For positive numbers x, y and z , the numerical value of the determinant $\begin{vmatrix} 1 & \log_x y & \log_x z \\ \log_y x & 1 & \log_y z \\ \log_z x & \log_z y & 1 \end{vmatrix}$ is-

- (A) 0 (B) $\log xyz$ (C) $\log(x + y + z)$ (D) $\log x \log y \log z$

DT0009

10. Let a determinant is given by $A = \begin{vmatrix} a & b & c \\ p & q & r \\ x & y & z \end{vmatrix}$ and suppose $A = 6$. If $B = \begin{vmatrix} p+x & q+y & r+z \\ a+x & b+y & c+z \\ a+p & b+q & c+r \end{vmatrix}$ then

- (A) $B = 6$ (B) $B = -6$ (C) $B = 12$ (D) $B = -12$

DT0010

11. The value of an odd order determinant in which $a_{ij} + a_{ji} = 0 \forall i, j$ is -

- (A) perfect square (B) negative (C) ± 1 (D) 0

DT0011

12. If $S_r = \begin{vmatrix} 2r & x & n(n+1) \\ 6r^2-1 & y & n^2(2n+3) \\ 4r^3-2nr & z & n^3(n+1) \end{vmatrix}$, then $\sum_{r=1}^n S_r$ does not depend on -

- (A) x (B) y (C) n (D) all of these

DT0012

13. If $a, b, c > 0$ and $x, y, z \in \mathbb{R}$, then the determinant $\begin{vmatrix} (a^x + a^{-x})^2 & (a^x - a^{-x})^2 & 1 \\ (b^y + b^{-y})^2 & (b^y - b^{-y})^2 & 1 \\ (c^z + c^{-z})^2 & (c^z - c^{-z})^2 & 1 \end{vmatrix}$ is equal to -

- (A) $a^x b^y c^x$ (B) $a^{-x} b^{-y} c^{-z}$ (C) $a^{2x} b^{2y} c^{2z}$ (D) zero

DT0013

14. If a, b, c are sides of a scalene triangle, then the value of $\begin{vmatrix} a & b & c \\ b & c & a \\ c & a & b \end{vmatrix}$ is :

(A) non-negative (B) negative (C) positive (D) non-positive

DT0014

15. The value of k for which the set of equations $3x+ky-2z=0$, $x+ky+3z=0$ and $2x+3y-4z=0$ has a non-trivial solution is-

(A) 15 (B) 16 (C) $31/2$ (D) $33/2$

DT0015

16. If the system of linear equations

$$x_1 + 2x_2 + 3x_3 = 6$$

$$x_1 + 3x_2 + 5x_3 = 9$$

$$2x_1 + 5x_2 + ax_3 = b$$

is consistent and has infinite number of solutions, then :-

(A) $a \in \mathbb{R} - \{8\}$ and $b \in \mathbb{R} - \{15\}$ (B) $a = 8$, b can be any real number
(C) $a = 8$, $b = 15$ (D) $b = 15$, a can be any real number

DT0016

17. Consider the system of equations : $x + ay = 0$, $y + az = 0$ and $z + ax = 0$. Then the set of all real values of ' a ' for which the system has a unique solution is :

(A) $\{1, -1\}$ (B) $\mathbb{R} - \{-1\}$ (C) $\{1, 0, -1\}$ (D) $\mathbb{R} - \{1\}$

DT0017

18. Let a, b, c be any real numbers. Suppose that there are real numbers x, y, z not all zero such that $x = cy + bz$, $y = az + cx$ and $z = bx + ay$, then $a^2 + b^2 + c^2 + 2abc$ is equal to

(A) 2 (B) -1 (C) 0 (D) 1

DT0018

19. Let $f(x) = \begin{vmatrix} 1 + \sin^2 x & \cos^2 x & 4 \sin 2x \\ \sin^2 x & 1 + \cos^2 x & 4 \sin 2x \\ \sin^2 x & \cos^2 x & 1 + 4 \sin 2x \end{vmatrix}$, then the maximum value of $f(x)$, is-

(A) 2 (B) 4 (C) 6 (D) 8

DT0019

20. If the determinant $\begin{vmatrix} a+p & \ell+x & u+f \\ b+q & m+y & v+g \\ c+r & n+z & w+h \end{vmatrix}$ splits into exactly K determinants of order 3, each

element of which contains only one term, then the value of K , is-

(A) 6 (B) 8 (C) 9 (D) 12

DT0020

21. Let $D_1 = \begin{vmatrix} a & b & a+b \\ c & d & c+d \\ a & b & a-b \end{vmatrix}$ and $D_2 = \begin{vmatrix} a & c & a+c \\ b & d & b+d \\ a & c & a+b+c \end{vmatrix}$ then the value of $\frac{D_1}{D_2}$ where $b \neq 0$ and

$ad \neq bc$, is

- (A) -2 (B) 0 (C) $-2b$ (D) $2b$

DT0021

22. If $a^2 + b^2 + c^2 = -2$ and $f(x) = \begin{vmatrix} 1+a^2x & (1+b^2)x & (1+c^2)x \\ (1+a^2)x & 1+b^2x & (1+c^2)x \\ (1+a^2)x & (1+b^2)x & 1+c^2x \end{vmatrix}$ then $f(x)$ is a polynomial of degree-

- (A) 0 (B) 1 (C) 2 (D) 3

DT0022

23. The number of real values of x satisfying $\begin{vmatrix} x & 3x+2 & 2x-1 \\ 2x-1 & 4x & 3x+1 \\ 7x-2 & 17x+6 & 12x-1 \end{vmatrix} = 0$ is -

- (A) 3 (B) 0 (C) 1 (D) infinite

DT0023

24. The determinant $\begin{vmatrix} \cos(\theta+\phi) & -\sin(\theta+\phi) & \cos 2\phi \\ \sin \theta & \cos \theta & \sin \phi \\ -\cos \theta & \sin \theta & \cos \phi \end{vmatrix}$ is -

- (A) 0 (B) independent of θ
(C) independent of ϕ (D) independent of θ & ϕ both

DT0024

25. If the system of equation, $a^2x - ay = 1 - a$ & $bx + (3 - 2b)y = 3 + a$ possess a unique solution $x = 1, y = 1$ than :

- (A) $a = 1; b = -1$ (B) $a = -1, b = 1$ (C) $a = 0, b = 0$ (D) none

DT0025

EXERCISE (O-2)

Multiple Correct Answer Type

1. The determinant $\begin{vmatrix} a^2 & a^2 - (b-c)^2 & bc \\ b^2 & b^2 - (c-a)^2 & ca \\ c^2 & c^2 - (a-b)^2 & ab \end{vmatrix}$ is divisible by -
- (A) $a + b + c$ (B) $(a + b)(b + c)(c + a)$
 (C) $a^2 + b^2 + c^2$ (D) $(a - b)(b - c)(c - a)$

DT0026

2. The value of θ lying between $-\frac{\pi}{4}$ & $\frac{\pi}{2}$ and $0 \leq A \leq \frac{\pi}{2}$ and satisfying the equation

$$\begin{vmatrix} 1 + \sin^2 A & \cos^2 A & 2 \sin 4\theta \\ \sin^2 A & 1 + \cos^2 A & 2 \sin 4\theta \\ \sin^2 A & \cos^2 A & 1 + 2 \sin 4\theta \end{vmatrix} = 0$$
 are -

- (A) $A = \frac{\pi}{4}, \theta = -\frac{\pi}{8}$ (B) $A = \frac{3\pi}{8} = \theta$
 (C) $A = \frac{\pi}{5}, \theta = -\frac{\pi}{8}$ (D) $A = \frac{\pi}{6}, \theta = \frac{3\pi}{8}$

DT0027

3. Which of the following determinant(s) vanish(es) ?

(A) $\begin{vmatrix} 1 & bc & bc(b+c) \\ 1 & ca & ca(c+a) \\ 1 & ab & ab(a+b) \end{vmatrix}$ (B) $\begin{vmatrix} 1 & ab & \frac{1}{a} + \frac{1}{b} \\ 1 & bc & \frac{1}{b} + \frac{1}{c} \\ 1 & ca & \frac{1}{c} + \frac{1}{a} \end{vmatrix}$

(C) $\begin{vmatrix} 0 & a-b & a-c \\ b-a & 0 & b-c \\ c-a & c-b & 0 \end{vmatrix}$ (D) $\begin{vmatrix} \log_x xyz & \log_x y & \log_x z \\ \log_y xyz & 1 & \log_y z \\ \log_z xyz & \log_z y & 1 \end{vmatrix}$

DT0028

4. The determinant $\begin{vmatrix} a & b & a\alpha + b \\ b & c & b\alpha + c \\ a\alpha + b & b\alpha + c & 0 \end{vmatrix}$ is equal to zero, if -
- (A) a, b, c are in AP (B) a, b, c are in GP
 (C) α is a root of the equation $ax^2 + bx + c = 0$ (D) $(x - \alpha)$ is a factor of $ax^2 + 2bx + c$

DT0029

5. System of linear equations in x, y, z

$$2x + y + z = 1$$

$$x - 2y + z = 2$$

$$3x - y + 2z = 3 \text{ have infinite solutions which}$$

(A) can be written as $(-3\lambda - 1, \lambda, 5\lambda + 3) \forall \lambda \in \mathbb{R}$

(B) can be written as $(3\lambda - 1, -\lambda, -5\lambda + 3) \forall \lambda \in \mathbb{R}$

(C) are such that every solution satisfy $x - 3y + 1 = 0$

(D) are such that none of them satisfy $5x + 3z = 1$

DT0030

6. System of equation $x + y + az = b$, $2x + 3y = 2a$ & $3x + 4y + a^2z = ab + 2$ has

(A) unique solution when $a \neq 0$, $b \in \mathbb{R}$

(B) no solution when $a = 0$, $b = 1$

(C) infinite solution when $a = 0$, $b = 2$

(D) infinite solution when $a = 1$, $b \in \mathbb{R}$

DT0031

7. If $\begin{vmatrix} a_1 + b_1x & a_1x + b_1 & c_1 \\ a_2 + b_2x & a_2x + b_2 & c_2 \\ a_3 + b_3x & a_3x + b_3 & c_3 \end{vmatrix} = 0$, then possible conditions is/are -

(A) $x = 1 \forall a_i, b_i$, where $1 \leq i, j \leq 3$

(B) $x = -1 \forall a_i, b_i$, where $1 \leq i, j \leq 3$

(C) $\begin{vmatrix} a_1 & b_1 & c_1 \\ a_2 & b_2 & c_2 \\ a_3 & b_3 & c_3 \end{vmatrix} = 0$

(D) $x = \pm 2 \forall a_i, b_i$, where $1 \leq i, j \leq 3$

DT0095

8. $\begin{vmatrix} x^2 & (y+z)^2 & yz \\ y^2 & (x+z)^2 & zx \\ z^2 & (x+y)^2 & xy \end{vmatrix}$ is divisible by -

(A) $x^2 + y^2 + z^2$

(B) $x - y$

(C) $x - y - z$

(D) $x + y + z$

DT0096

9. If $\begin{vmatrix} \frac{a}{b} + \frac{a}{c} & \frac{b}{c} + \frac{b}{a} & 1 \\ \frac{b}{a} + \frac{b}{c} & \frac{c}{a} + \frac{c}{b} & 1 \\ \frac{c}{a} + \frac{c}{b} & \frac{a}{b} + \frac{a}{c} & 1 \end{vmatrix} = 0$, where $a, b, c \in \mathbb{R}^+$, then which of the following is necessarily true -

(A) $\frac{1}{a} + \frac{1}{b} + \frac{1}{c} = 0$

(B) $a^2 + b^2 + c^2 = ab + bc + ac$

(C) $a = b = c$

(D) $\frac{1}{a} + \frac{1}{b} + \frac{1}{c} = 1$

DT0097

10. The system of linear equations $x + y + z = 6$, $x + 2y + 3z = 14$ and $2x + 5y + pz = q$ have -
 (A) infinitely many solution when $p = 8$, $q = 36$ (B) unique solution when $p \neq 8$, $q \neq 36$
 (C) no solution when $p = 8$, $q \neq 36$ (D) atleast one solution for $q = 36$, $p \in \mathbb{R}$

DT0098

11. If a, b, c are in A.P. and α, β, γ are positive real numbers in G.P., then the equation $\begin{vmatrix} x+a & x^2+\log\alpha & k \\ x+b & x^2+\log\beta & k \\ x+c & x^2+\log\gamma & k \end{vmatrix} = 0$:-
 (A) is an identity (B) has a root $x = 1$
 (C) has a root $x = 0$ (D) has real & identical roots

DT0099

12. If α, β, γ satisfy the equation $\begin{vmatrix} x & 1 & 1 \\ 1 & x & 1 \\ 1 & 1 & x \end{vmatrix} = 0$, then

(A) $\alpha + \beta + \gamma = 0$ (B) $\alpha^2 + \beta^2 + \gamma^2 = 6$ (C) $\alpha^3 + \beta^3 + \gamma^3 = -6$ (D) $\alpha^4 + \beta^4 + \gamma^4 = 18$

DT0100

13. If the system of equations $x + y - 3 = 0$, $(1 + K)x + (2 + K)y - 8 = 0$ & $x - (1 + K)y + (2 + K) = 0$ is consistent then the value of K may be -
 (A) 1 (B) $\frac{3}{5}$ (C) $-\frac{5}{3}$ (D) 2

DT0101

14. If system of equation $a_1x + b_1y = c_1$ & $a_2x + b_2y = c_2$ (where $a_1, b_1, c_1, a_2, b_2, c_2 \neq 0$) has infinite solutions, then-

(A) $\frac{a_1}{a_2} = \frac{b_1}{b_2} = \frac{c_1}{c_2}$

(B) $\frac{a_1 + a_2}{a_1 - a_2} = \frac{b_1 + b_2}{b_1 - b_2} = \frac{c_1 + c_2}{c_1 - c_2}$

(C) the quadratic equations $a_1x^2 + b_1x + c_1 = 0$ & $a_2x^2 + b_2x + c_2 = 0$ have no common root

(D) system of equation $a_1^2a_2x + b_1^2b_2y = c_1^2c_2$ & $a_1a_2^2x + b_1b_2^2y = c_1c_2^2$ will also have infinite number of solutions

DT0102

15. If $\ell < m < n$, then $\begin{vmatrix} 1 & \ell & \ell^4 \\ 1 & m & m^4 \\ 1 & n & n^4 \end{vmatrix}$ will always be greater than -

(A) 3 (B) 6 (C) -1 (D) 0

DT0103

EXERCISE (O-3)

Linked Comprehension Type

Paragraph for Question 1 to 3

$$\text{Let } x, y, z \in \mathbb{R}^+ \text{ \& } D = \begin{vmatrix} x & x^3 & x^4 - 1 \\ y & y^3 & y^4 - 1 \\ z & z^3 & z^4 - 1 \end{vmatrix}$$

On the basis of above information, answer the following questions :

1. If $x \neq y \neq z$ & x, y, z are in GP and $D = 0$, then y is equal to -
(A) 1 (B) 2 (C) 4 (D) none of these

DT0104

2. If x, y, z are the roots of $t^3 - 21t^2 + bt - 343 = 0$, $b \in \mathbb{R}$, then D is equal to-
(A) 1 (B) 0 (C) dependent on x, y, z (D) data inadequate

DT0105

3. If $x \neq y \neq z$ & x, y, z are in A.P. and $D = 0$, then $2xyz + x^2z^2$ is equal to-
(A) 1 (B) 2 (C) 3 (D) none of these

DT0106

Paragraph for Question 4 to 6

$$\text{Let } \begin{vmatrix} 1 & 1 & 1 \\ 1 & 2 & 3 \\ \alpha & \beta & \gamma \end{vmatrix} = t, \text{ where } t \text{ is an even prime number \& } \alpha, \beta, \gamma \text{ are the integral roots of the equation}$$

$$x^3 - 14x^2 + Px - 36 = 0$$

On the basis of above information answer the following :

4. The value of P is-
(A) a rational number (B) a prime number
(C) an odd natural number (D) an even natural number
5. The value of $\alpha(\beta^2 + \gamma^2) + \beta(\gamma^2 + \alpha^2) + \gamma(\alpha^2 + \beta^2)$ is divisible by -
(A) 17 (B) 34 (C) 51 (D) 68

DT0107

6. Which of following statement is/are false -
(A) t is divisible by $(\alpha - \beta)$ (B) t is divisible by $(\beta - \gamma)$
(C) t is divisible by $(\gamma - \alpha)$ (D) $(\gamma - \alpha)$ is divisible by t

DT0108

DT0109

Paragraph for Question 7 & 8

$$\text{Let } f(x) = \begin{vmatrix} 1 & x-1 & x \\ 2(x-1) & (x-1)(x-2) & x(x-1) \\ 3(x-1)(x-2) & (x-1)(x-2)(x-3) & x(x-1)(x-2) \end{vmatrix} \quad \& \quad D_r = \begin{vmatrix} 1 & 10 & 2r \\ 70 & 17 & 3r+1 \\ 1 & 1 & 1 \end{vmatrix}.$$

On the basis of above information, answer the following questions :

7. The value of $f(50) - D_5$ is -

DT0110

8. $\sum_{r=1}^{10} D_r$ is

DT0111

Matrix Match Type

9. Consider a system of linear equations $a_i x + b_i y + c_i z = d_i$ (where $a_i, b_i, c_i \neq 0$ and $i = 1, 2, 3$) & (α, β, γ) is its unique solution, then match list-I with list-II

List-I

List-II

- | | |
|---|--------|
| (I) If $a_i = k$, $d_i = k^2$, ($k \neq 0$) and $\alpha + \beta + \gamma = 2$, then k is | (P) 1 |
| (II) If $a_i = d_i = k \neq 0$, then $\alpha + \beta + \gamma$ is | (Q) 2 |
| (III) If $a_i = k > 0$, $d_i = k + 1$, then $\alpha + \beta + \gamma$ can be | (R) 0 |
| (IV) If $a_i = k < 0$, $d_i = k + 1$, then $\alpha + \beta + \gamma$ can be | (S) 3 |
| | (T) -1 |

DT0032

- (A) I $\rightarrow$ P,Q; II $\rightarrow$ R; III $\rightarrow$ Q,S; IV $\rightarrow$ T
 (B) I $\rightarrow$ P; II $\rightarrow$ Q; III $\rightarrow$ R,S; IV $\rightarrow$ T
 (C) I $\rightarrow$ Q; II $\rightarrow$ P,R; III $\rightarrow$ S; IV $\rightarrow$ T
 (D) I $\rightarrow$ Q; II $\rightarrow$ P; III $\rightarrow$ Q,S; IV $\rightarrow$ R,T

10. Match the following for the system of linear equations

$$\lambda x + y + z = 1, \quad x + \lambda y + z = \lambda, \quad x + y + \lambda z = \lambda^2$$

Column-I

Column-II

- | | |
|---------------------------------------|---------------------------|
| (A) $\lambda = 1$ | (P) unique solution |
| (B) $\lambda \neq 1$ | (Q) infinite solutions |
| (C) $\lambda \neq 1, \lambda \neq -2$ | (R) no solution |
| (D) $\lambda = -2$ | (S) finite many solutions |

DT0112

EXERCISE (O-4)

Numerical Grid Type

1. Let a, b, c are the solutions of the cubic $x^3 - 5x^2 + 3x - 1 = 0$, then find the value of the determinant

$$\begin{vmatrix} a & b & c \\ a-b & b-c & c-a \\ b+c & c+a & a+b \end{vmatrix}.$$

DT0040

2. If $\Delta(x) = \begin{vmatrix} 0 & 2x-2 & 2x+8 \\ x-1 & 4 & x^2+7 \\ 0 & 0 & x+4 \end{vmatrix}$ and $f(x) = \sum_{j=1}^3 \sum_{i=1}^3 a_{ij}c_{ij}$, where a_{ij} is the element of i^{th} and j^{th} column

in $\Delta(x)$ and c_{ij} is the cofactor $a_{ij} \forall i$ and j , then find the greatest value of $f(x)$, where $x \in [-3, 18]$

DT0041

3. If the equations $a(y+z) = x$, $b(z+x) = y$, $c(x+y) = z$ (where $a, b, c \neq -1$) have nontrivial solutions, then find the value of $\frac{1}{1+a} + \frac{1}{1+b} + \frac{1}{1+c}$.

DT0053

4. Find the sum of all positive integral values of a for which every solution to the system of equation $x + ay = 3$ and $ax + 4y = 6$ satisfy the inequalities $x > 1$, $y > 0$.

DT0062

5. For a determinant Δ of order 3, the element a_{ij} is defined as $a_{ij} = \tan^{-1}(\tan(i-j)) \forall i, j$, then the value of Δ is equal to (where 'i' represents row and 'j' represents column)

DT0113

6. The number of triplets (α, β, γ) satisfying the following constraints

$$2\alpha - \beta + 3\gamma = 4$$

$$\alpha + \beta - 3\gamma = -1$$

$$5\alpha - \beta + 3\gamma = 7$$

$$\alpha\beta\gamma \leq 0$$

$$\& \quad \alpha, \beta, \gamma \in I$$

DT0114

7. Let $a, b, c, \ell, m, n \in \mathbb{R}$ such that $a\ell + bm + cn = 0$, $b\ell + cm + an = 0$, $c\ell + am + bn = 0$. If a, b & c are distinct & $f(x) = ax^3 + bx^2 + cx + 5$, then the value of $f(1)$ is

DT0115

8. If $\begin{vmatrix} \sin^2 2x & \cos 2x & 4\sin^2 \frac{x}{2} \\ \tan^2 \frac{x}{2} & \cos^2 x & -\sin^2 x \\ -2\cos 4x & \tan^2 \frac{x}{2} & \sin 4x \end{vmatrix} = a_0 + a_1(\cos x) + a_2(\cos^2 x) + \dots + a_n(\cos^n x)$, then a_0 is -

DT0116

9. If x, y, z are distinct digits ($0 \leq x, y, z \leq 9$) & the minimum possible value of $\begin{vmatrix} z & 9y & x \\ z & y & 9x \\ 9z & y & x \end{vmatrix}$ is λ

then $\frac{\lambda}{83700}$ is

(where $9x, 9y$ & $9z$ are two digits number)

DT0117

10. $\begin{vmatrix} a & b-c & c+b \\ a+c & b & c-a \\ a-b & b+a & c \end{vmatrix} = \begin{vmatrix} b-c & b & b+a \\ a & -a-c & b-a \\ c+b & c-a & c \end{vmatrix}$ and $a > 0, b > 0, c > 0$, then the minimum value of $\frac{c^2}{ab}$ is

DT0118

EXERCISE (JM)

1. The number of values of k , for which the system of equations : [JEE(Main)-2013]

$$(k + 1)x + 8y = 4k, \quad kx + (k + 3)y = 3k - 1 \text{ has no solution, is -}$$

- (1) infinite (2) 1 (3) 2 (4) 3

DT0069

2. If $\alpha, \beta \neq 0$, and $f(n) = \alpha^n + \beta^n$ and $\begin{vmatrix} 3 & 1+f(1) & 1+f(2) \\ 1+f(1) & 1+f(2) & 1+f(3) \\ 1+f(2) & 1+f(3) & 1+f(4) \end{vmatrix} = K(1 - \alpha)^2 (1 - \beta)^2 (\alpha - \beta)^2$, then

K is equal to :

[JEE(Main)-2014]

- (1) $\alpha\beta$ (2) $\frac{1}{\alpha\beta}$ (3) 1 (4) -1

DT0070

3. The set of all values of λ for which the system of linear equations :
 $2x_1 - 2x_2 + x_3 = \lambda x_1, 2x_1 - 3x_2 + 2x_3 = \lambda x_2, -x_1 + 2x_2 = \lambda x_3$ has a non-trivial solution

[JEE(Main)-2015]

- (1) contains two elements (2) contains more than two elements
 (3) is an empty set (4) is a singleton

DT0071

4. The system of linear equations $x + \lambda y - z = 0, \lambda x - y - z = 0, x + y - \lambda z = 0$ has a non-trivial solution for : [JEE(Main)-2016]

- (1) exactly three values of λ . (2) infinitely many values of λ .
 (3) exactly one value of λ . (4) exactly two values of λ .

DT0072

5. If S is the set of distinct values of ' b ' for which the following system of linear equations

$$\begin{aligned} x + y + z &= 1 \\ x + ay + z &= 1 \\ ax + by + z &= 0 \end{aligned}$$

has no solution, then S is :

[JEE(Main)-2017]

- (1) a singleton (2) an empty set
 (3) an infinite set (4) a finite set containing two or more elements

DT0073

6. If $\begin{vmatrix} x-4 & 2x & 2x \\ 2x & x-4 & 2x \\ 2x & 2x & x-4 \end{vmatrix} = (A + Bx)(x - A)^2$, then the ordered pair (A, B) is equal to :

[JEE(Main)-2018]

- (1) $(-4, 3)$ (2) $(-4, 5)$ (3) $(4, 5)$ (4) $(-4, -5)$

DT0074

7. If the system of linear equations
- $$\begin{aligned}x + ky + 3z &= 0 \\3x + ky - 2z &= 0 \\2x + 4y - 3z &= 0\end{aligned}$$

has a non-zero solution (x, y, z) , then $\frac{xz}{y^2}$ is equal to :

[JEE(Main)-2018]

- (1) 10 (2) -30 (3) 30 (4) -10

DT0075

8. If the system of linear equations
- $$\begin{aligned}x - 4y + 7z &= g \\3y - 5z &= h \\-2x + 5y - 9z &= k\end{aligned}$$

is consistent, then :

[JEE(Main) 2019]

- (1) $g + h + k = 0$ (2) $2g + h + k = 0$
(3) $g + h + 2k = 0$ (4) $g + 2h + k = 0$

DT0076

9. Let $d \in \mathbb{R}$, and $A = \begin{bmatrix} -2 & 4+d & (\sin \theta) - 2 \\ 1 & (\sin \theta) + 2 & d \\ 5 & (2\sin \theta) - d & (-\sin \theta) + 2 + 2d \end{bmatrix}$, $\theta \in [0, 2\pi]$. If the minimum value of $\det(A)$

is 8, then a value of d is :

[JEE(Main) 2019]

- (1) -7 (2) $2(\sqrt{2} + 2)$ (3) -5 (4) $2(\sqrt{2} + 1)$

DT0077

10. Let $a_1, a_2, a_3, \dots, a_{10}$ be in G.P. with $a_i > 0$ for $i = 1, 2, \dots, 10$ and S be the set of pairs (r, k) , $r, k \in \mathbb{N}$ (the

set of natural numbers) for which $\begin{vmatrix} \log_e a_1^r a_2^k & \log_e a_2^r a_3^k & \log_e a_3^r a_4^k \\ \log_e a_4^r a_5^k & \log_e a_5^r a_6^k & \log_e a_6^r a_7^k \\ \log_e a_7^r a_8^k & \log_e a_8^r a_9^k & \log_e a_9^r a_{10}^k \end{vmatrix} = 0$. Then the number of elements

in S , is :

[JEE(Main) 2019]

- (1) Infinitely many (2) 4 (3) 10 (4) 2

DT0078

11. The set of all values of λ for which the system of linear equations.

$$\begin{aligned}x - 2y - 2z &= \lambda x \\x + 2y + z &= \lambda y \\-x - y &= \lambda z\end{aligned}$$

has a non-trivial solution.

[JEE(Main) 2019]

- (1) contains more than two elements
(2) is a singleton
(3) is an empty set
(4) contains exactly two elements

DT0079

12. If the system of linear equations

$$2x + 2ay + az = 0$$

$$2x + 3by + bz = 0$$

$$2x + 4cy + cz = 0,$$

[JEE(Main) 2020]

where $a, b, c \in \mathbb{R}$ are non-zero and distinct; has a non-zero solution, then :

(1) a, b, c are in A.P.

(2) $a + b + c = 0$

(3) a, b, c are in G.P.

(4) $\frac{1}{a}, \frac{1}{b}, \frac{1}{c}$ are in A.P.

DT0080

13. The system of linear equations

$$\lambda x + 2y + 2z = 5$$

$$2\lambda x + 3y + 5z = 8$$

$$4x + \lambda y + 6z = 10$$

[JEE(Main) 2020]

(1) infinitely many solutions when $\lambda = 2$

(2) a unique solution when $\lambda = -8$

(3) no solution when $\lambda = 8$

(4) no solution when $\lambda = 2$

DT0081

14. For which of the following ordered pairs (μ, δ) , the system of linear equations

$$x + 2y + 3z = 1$$

$$3x + 4y + 5z = \mu$$

$$4x + 4y + 4z = \delta$$

is inconsistent ?

[JEE(Main) 2020]

(1) (1,0)

(2) (4,6)

(3) (3,4)

(4) (4,3)

DT0082

15. Let $a - 2b + c = 1$. If $f(x) = \begin{vmatrix} x+a & x+2 & x+1 \\ x+b & x+3 & x+2 \\ x+c & x+4 & x+3 \end{vmatrix}$, then :

[JEE(Main) 2020]

(1) $f(-50) = 501$

(2) $f(-50) = -1$

(3) $f(50) = 1$

(4) $f(50) = -501$

DT0083

16. Let α, β, γ be the real roots of the equation, $x^3 + ax^2 + bx + c = 0$, ($a, b, c \in \mathbb{R}$ and $a, b \neq 0$). If the system of equations (in u, v, w) given by $\alpha u + \beta v + \gamma w = 0$, $\beta u + \gamma v + \alpha w = 0$; $\gamma u + \alpha v + \beta w = 0$ has

non-trivial solution, then the value of $\frac{a^2}{b}$ is

[JEE(Main) 2021]

(1) 5

(2) 3

(3) 1

(4) 0

DT0119

17. Let the system of linear equations

$$4x + \lambda y + 2z = 0$$

$$2x - y + z = 0$$

$$\mu x + 2y + 3z = 0, \lambda, \mu \in \mathbb{R}.$$

has a non-trivial solution. Then which of the following is true ?

[JEE(Main) 2021]

(1) $\mu = 6, \lambda \in \mathbb{R}$

(2) $\lambda = 2, \mu \in \mathbb{R}$

(3) $\mu = 3, \mu \in \mathbb{R}$

(4) $\mu = -6, \lambda \in \mathbb{R}$

DT0120

18. The number of real values λ , such that the system of linear equations

$$2x - 3y + 5z = 9$$

$$x + 3y - z = -18$$

$$3x - y + (\lambda^2 - |\lambda|)z = 16$$

has no solution, is :-

[JEE(Main) 2022]

(A) 0

(B) 1

(C) 2

(D) 4

DT0121

19. If the system of equations

$$x + y + z = 6$$

$$2x + 5y + \alpha z = \beta$$

$$x + 2y + 3z = 14$$

has infinitely many solutions, then $\alpha + \beta$ is equal to :

[JEE(Main) 2022]

(A) 8

(B) 36

(C) 44

(D) 48

DT0122

EXERCISE (JA)

1. The number of all possible values of θ , where $0 < \theta < \pi$, for which the system of equations

$$(y + z)\cos 3\theta = (xyz)\sin 3\theta$$

$$x \sin 3\theta = \frac{2\cos 3\theta}{y} + \frac{2\sin 3\theta}{z}$$

$$(xyz)\sin 3\theta = (y + 2z)\cos 3\theta + y\sin 3\theta$$

have a solution (x_0, y_0, z_0) with $y_0 z_0 \neq 0$, is

[JEE 2010, 3]

DT0084

2. Which of the following values of α satisfy the equation
- $$\begin{vmatrix} (1+\alpha)^2 & (1+2\alpha)^2 & (1+3\alpha)^2 \\ (2+\alpha)^2 & (2+2\alpha)^2 & (2+3\alpha)^2 \\ (3+\alpha)^2 & (3+2\alpha)^2 & (3+3\alpha)^2 \end{vmatrix} = -648\alpha ?$$

[JEE(Advanced)-2015, 4M, -2M]

(A) -4

(B) 9

(C) -9

(D) 4

DT0085

3. The total number of distinct $x \in \mathbb{R}$ for which
- $$\begin{vmatrix} x & x^2 & 1+x^3 \\ 2x & 4x^2 & 1+8x^3 \\ 3x & 9x^2 & 1+27x^3 \end{vmatrix} = 10$$

[JEE(Advanced)-2016, 3(0)]

DT0086

4. Let $a, \lambda, m \in \mathbb{R}$. Consider the system of linear equations

$$ax + 2y = \lambda$$

$$3x - 2y = \mu$$

Which of the following statement(s) is(are) correct ?

- (A) If $a = -3$, then the system has infinitely many solutions for all values of λ and μ
 (B) If $a \neq -3$, then the system has a unique solution for all values of λ and μ
 (C) If $\lambda + \mu = 0$, then the system has infinitely many solutions for $a = -3$
 (D) If $\lambda + \mu \neq 0$, then the system has no solution for $a = -3$ [JEE(Advanced)-2016, 4(-2)]

DT0087

5. Let p, q, r be nonzero real numbers that are, respectively, the 10^{th} , 100^{th} and 1000^{th} terms of a harmonic progression. Consider the system of linear equations

$$x + y + z = 1$$

$$10x + 100y + 1000z = 0$$

$$qr x + pr y + pq z = 0.$$

[JEE(Advanced)-2022]

List-I

List-II

- | | |
|--|---|
| (I) If $\frac{q}{r} = 10$, then the system of linear equations has | (P) $x = 0, y = \frac{10}{9}, z = -\frac{1}{9}$ as a solution |
| (II) If $\frac{p}{r} \neq 100$, then the system of linear equations has | (Q) $x = \frac{10}{9}, y = -\frac{1}{9}, z = 0$ as a solution |
| (III) If $\frac{p}{q} \neq 10$, then the system of linear equations has | (R) infinitely many solutions |
| (IV) If $\frac{p}{q} = 10$, then the system of linear equations has | (S) no solution |
| | (T) at least one solution |

The correct option is:

- (A) (I) $\rightarrow$ (T); (II) $\rightarrow$ (R); (III) $\rightarrow$ (S); (IV) $\rightarrow$ (T)
 (B) (I) $\rightarrow$ (Q); (II) $\rightarrow$ (S); (III) $\rightarrow$ (S); (IV) $\rightarrow$ (R)
 (C) (I) $\rightarrow$ (Q); (II) $\rightarrow$ (R); (III) $\rightarrow$ (P); (IV) $\rightarrow$ (R)
 (D) (I) $\rightarrow$ (T); (II) $\rightarrow$ (S); (III) $\rightarrow$ (P); (IV) $\rightarrow$ (T)

DT0123

ANSWER KEY

Do yourself -1

1. minors : 4, -1, -4, 4 ; cofactors : -4, -1, 4, 4 2. -98 3. B 4. 0
5. C 6. A 7. C 8. D 9. D 10. A

Do yourself -2

2. C 3. 0 4. A 5. D

Do yourself -3

1. 2 2. 0 3. C 4. B 5. 0.00

Do yourself -4

2. $x = -1, 2$ 3. A 4. C 5. B

Do yourself -5

1. $\begin{vmatrix} 1 & 1 & 1 \\ 1 & \alpha & \beta \\ 1 & \beta & \alpha \end{vmatrix}$ 2. D 3. A 4. C 5. A

Do yourself -6

1. C 2. 0 4. A 5. B

Do yourself -7

1. infinite solutions 2. A 3. A 4. D 5. D 6. B
7. C 8. B 9. B

EXERCISE # O-1

Que.	1	2	3	4	5	6	7	8	9	10
Ans.	D	A	C	A	B	D	B	C	A	C
Que.	11	12	13	14	15	16	17	18	19	20
Ans.	D	D	D	B	D	C	B	D	C	B
Que.	21	22	23	24	25					
Ans.	A	C	D	B	A					

EXERCISE # O-2

Que.	1	2	3	4	5	6	7	8	9	10
Ans.	A,C,D	A,B,C,D	A,B,C,D	B,D	A,B,D	B,C,D	A,B,C	A,B,D	B,C	A,B,C,D
Que.	11	12	13	14	15					
Ans.	A,B,C	A,B,C,D	A,C	A,B,D	C,D					

EXERCISE # O-3

Que.	1	2	3	4	5	6	7	8	9	
Ans.	A	B	C	A,C	A,B	A,B,C	9.00	575.00	D	
Que. 10	A	B	C	D						
	Q	P,R	P	R						

EXERCISE # O-4

Que.	1	2	3	4	5	6	7	8	9	10
Ans.	80	0	2	4	0	1	5	0	9	4

EXERCISE # JEE-MAIN

Que.	1	2	3	4	5	6	7	8	9	10
Ans.	2	3	1	1	1	2	1	2	3	1
Que.	11	12	13	14	15	16	17	18	19	
Ans.	2	4	4	4	3	2	1	C	C	

EXERCISE # JEE-ADVANCED

Que.	1	2	3	4	5					
Ans.	3	B,C	2	B,C,D	B					